M T W T F S

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Topic: Backup and Recovery

Backup:

⇒ The process of periodically taking a copy of the database and log file onto offline storage media.

⇒ A backup is a copy of data.

Need of Taking backup

1. **Instance failure:**

⇒ Instance shuts down without synchronizing all the database files to the same system change number.

⇒ A few causes for instance failure:

- A power outage
- A server hardware failure
- Failure of oracle background
- Processes.
- Emergency shutdown procedures

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Solution:

STARTUP database and let oracle automatically perform inst. recovery using the online redo logs and undo the data in the undo tablespace.

2. User error failures:

- Inadvertently delete or modify data in tables or drop an index

3. media failures:

- The loss of one or more database files.
- The database file can be lost or corrupted for a number of reasons:

⇒ Failure of disk drive

⇒ Failure of disk controller

⇒ Inadvertent deletion or corruption of a database file.

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Types of backup

(1) Hot / online backups:

- Backup of one or more database files while the database is open.
- Incosistent
- In 24 x 7 environment when used.
- Database should be in Archivelog mode.
- steps involve:

⇒ Setting tablespace into a each state.

⇒ Backing up backup datafiles

⇒ Then restoring the tablespace to normal.

- Files to backup while open:

⇒ All datafiles

⇒ All archived redo log files

⇒ one control file

⇒ server parameter file

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(2) Cold / offline backups:

⇒ Backing up the database when it is closed (shutdown state).

⇒ Consistent

○ No need of recovery because the data is already consistent.

User-managed backups

⇒ This strategy is to make periodic backups of datafiles and archived logs with OS commands.

(i) Physical backups:

○ Backup of physical file structure (datafiles, control files...etc)

○ Involves copying the files that make up the database.

○ These backups also calls file system backups, use OS file backup commands.

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Types

Offline / cold / consistent

↳ occurs when database is shut down normally

↳ backup all control files, datafiles, server files

↳ Provide a complete snapshot of the db at the time of shutdown.

↳ Don't backup redo log files, bcz they should protected through multiplexing / mirroring

online / Hot / Inconsistent

↳ need archived and redo logs for recovery

(ii) Logical backups:

⇒ Backups of logical objects such as tables, views

○ Export / import (Dump) utility is used for these backups.

⇒ involves reading a set of data records and writing them to a file, independent of their physical location.

Data pump export (expdp)

- o Reads database record information, including data dictionary information, and writes them to xml file as export dump file.
- o can export the entire database, specific users, tablespaces or individual tables

Data pump import (impdp)

- o Reads the export dump file created by data pump export.
- o Executes the command in the dump file to recreate the objects and insert data.

Full and incremental backup

Full backups:

⇒ Include all blocks of datafiles, a complete copy.

⇒ Back up of all datafiles and whole tablespaces.

Incremental backup

⇒ Level 0 is a full backup.

⇒ Level 1 changes since the last backup.

⇒ Saves space and resources.

Differential incremental backup

↳ backup all data blocks that have changed b/w level 0 and level 1

Commulative incremental backup

↳ Backup all the blocks till the level 1.

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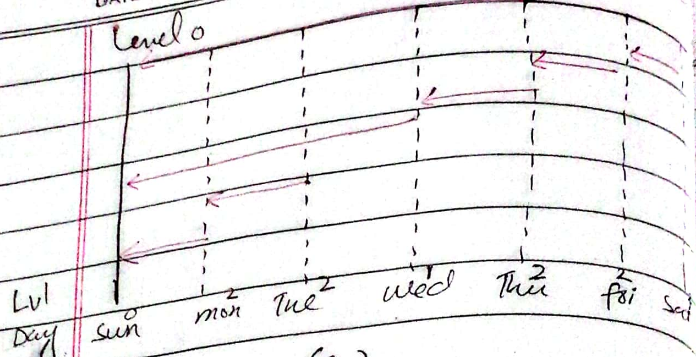
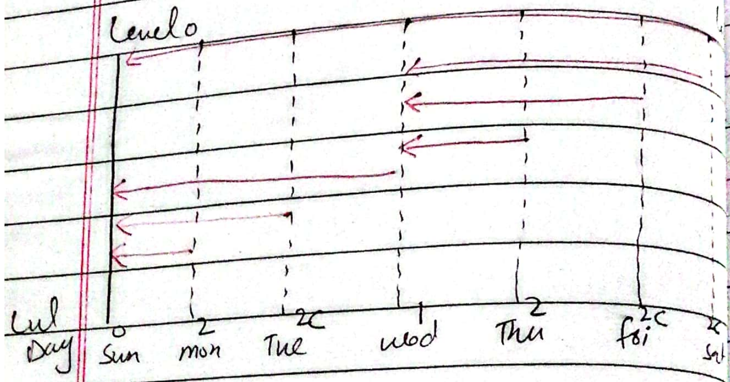


(a)



Basic backup methodology

- Identify all datafiles
- Use an OS Command
- Use a SQL statement

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RMAN

Definition:

⟹ Oracle provided utility for backing-up, restoring and recovering oracle databases.

⟹ Does not require a separate installation.

⟹ RMAN is a backup and recovery manager.

⟹ it provides database backup, restore and recovery capabilities.

RMAN features

⟹ Compress backups of datafiles so that only those data blocks that have been written to are included in a backup.

Store frequently executed backup and recovery operations in scripts.

Perform crosschecks

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⇒ Recovery catalog automates both restore and recovery operations

RMAN components

RMAN executable:
can be invoked from a command line prompt.
Found in $ORACLE_HOME.

Target database:
- The database to be backed up by RMAN.

- information about backups is stored in the control file or separate file.

Flash recovery area:
- Simplifies disk based backup
- DBAs specify location and disk space limit.

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media management Software:
⇒ Supports tape and other non-disk based backups.
⇒ Work with third-party media management software.

Advantages of RMAN:

Skipping unused blocks:
- Does not back up blocks that have never been written to.

Backup compression:
- RMAN uses an oracle specific compression algorithm
- Save space and media use

Open database backups:
- RMAN can backup tablespaces without additional commands.

True incremental backup:
- RMAN only back up blocks that have changed since last back up.

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Block level recovery:
- RMAN can restore or repair corrupted blocks.

Multiple I/O channels:
- RMAN can perform concurrent operations.

Platform independence:
- RMAN consistent across different platforms.

Tape manager support:
- support major enterprise back

Cataloging:
- RMAN keeps a record of all backups.

Scripting capabilities:
- RMAN scripts can stored and scheduled easily.

Encrypted backups:
- RMAN supports encrypted backups.

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Traditional backup limitations:

Non-database files: RMAN does not back up password files or configuration files like tnsnames.ora. These can be backed up using traditional methods.

Q. What kinds of backup perform by RMAN?

⇒ all Datafiles and control files
⇒ server parameter files
⇒ Archived redo logs
⇒ RMAN backups

Consistent backup

- All datafiles have the same SCN and include all changes from redo log.
- Taken when database is shut down normally.

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Inconsistent backup:

- Taken while database is open, needing archived online redo logs for recovery.

Backup set:

- RMAN can store backup data in a logical structure called a backup set, which is the smallest unit of an RMAN backup.
- Backup set contains the data from one or more datafiles, archived redo logs or control files or server parameter files.

⇒ Usually contain more than one file.
⇒ can be written on disk or tape
⇒ required to extract files for restoration.

Backup piece:

A backup set contains one or binary files in an RMAN-specific format. This file is known as backup piece.

A file in backup set.

⇒ A backup set contains multiple datafiles. blocks.

Full & Incremental backups:

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compressed backups:

Reduces disk space or tape space needed. Automatically decompressed by RMAN during recovery.

Image copies:

- are exact byte-for-byte copies of file.

Create an image copy by copying a file at the os level. image copies created through RMAN are recorded in the RMAN repository.

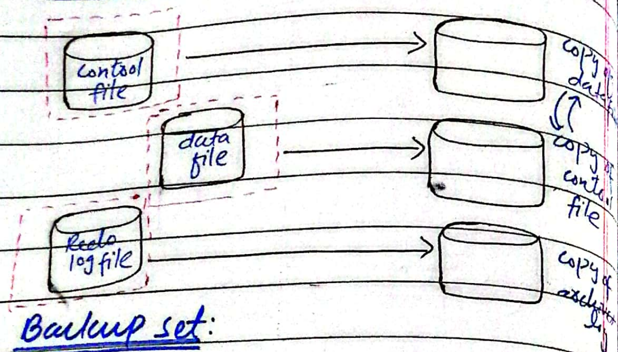
Crosscheck:

- This command is use to check the status of backupsets and image copies.

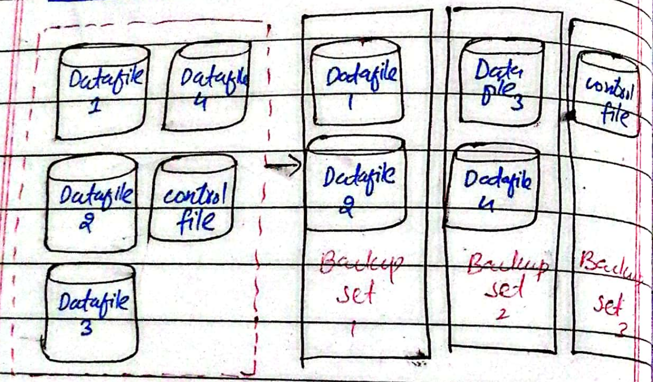
List expired backups:

- Used to view expired backups.

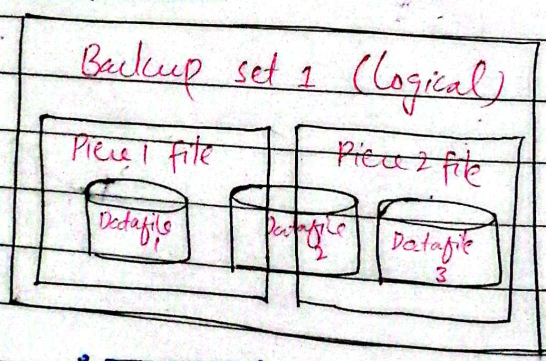
Image files



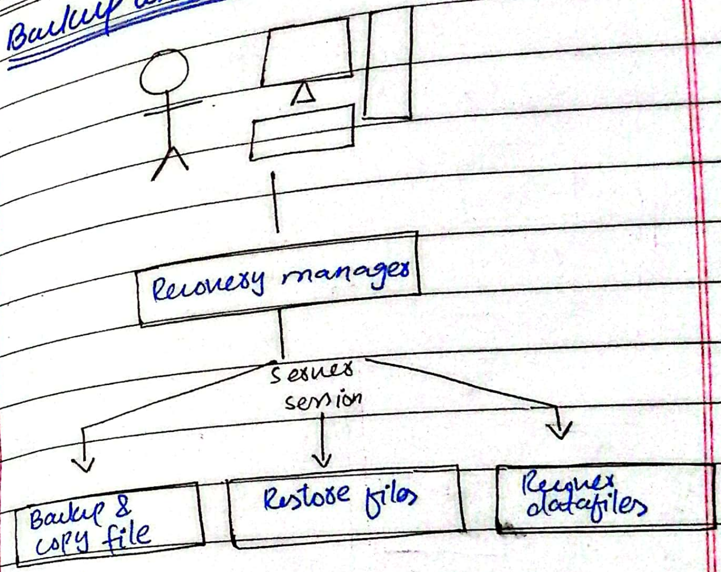
Backup set:



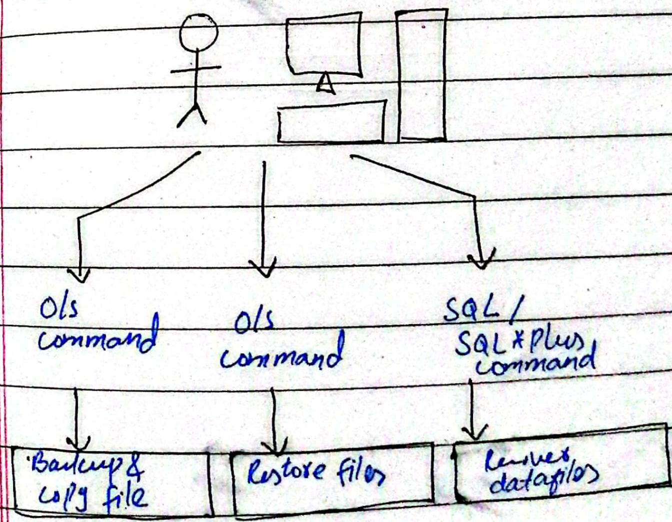
Backup Piece:



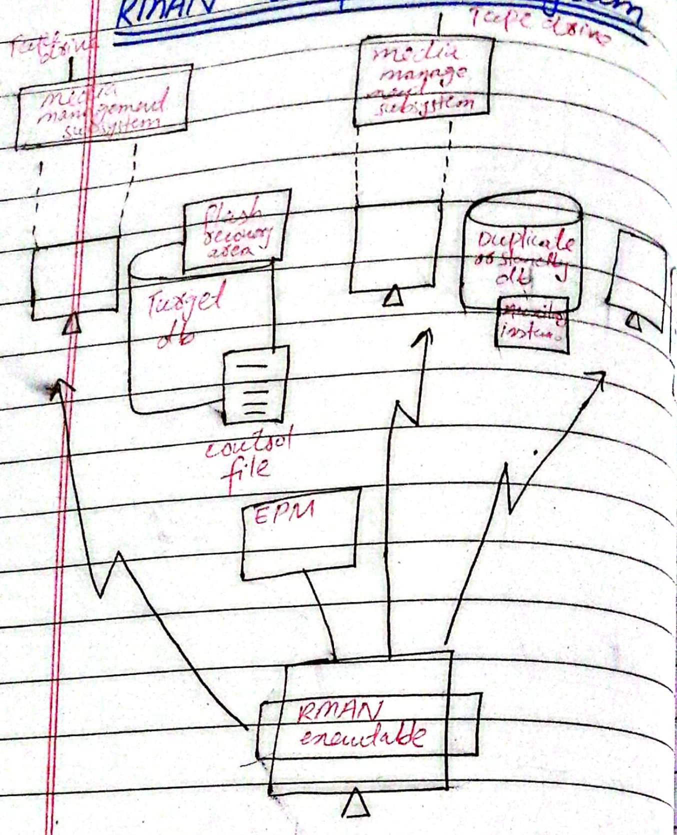
Backup and recovery with recovery manager:



O/s backup and recovery:



RMAN Complete Diagram



RMAN Recovery operations

Granularity of recovery: RMAN can restore and recover at different levels - from individual block to entire database

Database Availability:

Many recovery operations can be done while the database is still open and in use.

Validation methods:

RMAN can validate restore operations without actually recovering the datafiles.

Block media recovery

Purpose: Used when only few blocks in the database are corrupted, saving time and resources compared to full datafile recovery.

Availability: only accessible through RMAN.

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Steps:

⇒ Identify the data file & blocks numb

⇒ Use RMAN command to recover the block 'RMAN> recover datafile 6 block 403'

RMAN control file autobackup

- CONFIGURE CONTROLFILE AUTOBACKUP
- When enabled, RMAN automatically performs a control file autobackup after BCAK or COPY command.

Restoring a control file

- Loss of control file:
  if all control files are lost, restore using RMAN.
- with recovery catalog:
  Use: 'RMAN> restore catalogfile
- without recovery catalog:
  specify the location of the control file backup

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After restoration:
  Perform complete media recovery after restoration & open the db with 'resetlog'.

Archived redo log backups

⇒ Archived redo logs are essential for recovering an inconsistent backup.

- BACKUP ARCHIVELOG

Restoring a Tablespace

Scenario:

Disk containing datafiles for a tablespace fails or becomes corrupted.

Check datafile status:

Use $DATAFILE_HEADER to identify which datafile need recovery.

Recover the affected tablespace.

⇒ Recover Tablespace uses

Ex:

- Force tablespace offline.
- Restore tablespace
- Recover tablespace
- Bring tablespace online

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Restoring an entire database

Event:
Loss of all datafiles.

Preparation:
- Ensure control file and redo logs are available
- Adjust parameter file

Steps to restore:
- Start RMAN and mount the database
- Restore database
- Recover database
- open database

Restoring backups of the damaged or missing files:

⟹ Determine which datafiles to recover:

Select * FROM V$RECOVERY

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Query V$DATAFILE and V$TABLESPACE to obtain filenames and tablespaces names for datafiles requiring recovery.

⟹ In case of media failure, indicate the new locations of datafiles to the control files.

Recreating datafiles when backups are unavailable

- All archived log files written after the creation of the original datafile must be available.
- The control file contains the name of the damaged file.
- Create a new, empty datafile to replace a damaged datafile that has no corresponding backup.
- Perform media recovery on the empty datafile.

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RMAN recovery Techniques

⇒ Automated file creation during recovery

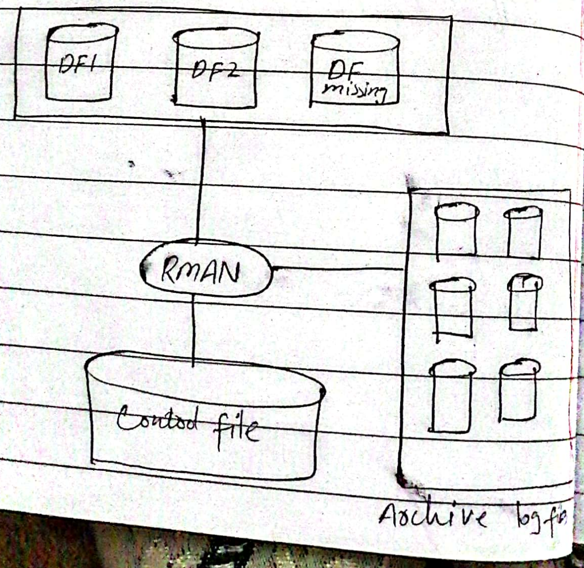
⇒ Simplified recovery through Restlogs.

⇒ Change-aware incremental backups

⇒ Automated disk-based backup and recovery.

⇒ RMAN database dropping and deregistration.

Automated file creation during recovery:



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Recovery using Flashback functions

- Flashback is an oracle database facility to quickly move an entire database or a table back to a prior state for recovery purposes.
- Flashback first introduced in 9i

FRA (flasback recovery area)

(directory on disk or ASM diskgroup)

⇒ Storage area that enables database flash backup and recovery operations. Provides centralized location

- first introduced with oracle 10g.

⇒ Related parameters:

- db_recovery_file_dest - specify location
- db_recovery_file_dest_size → specify the size
- db_flashback_retention_target

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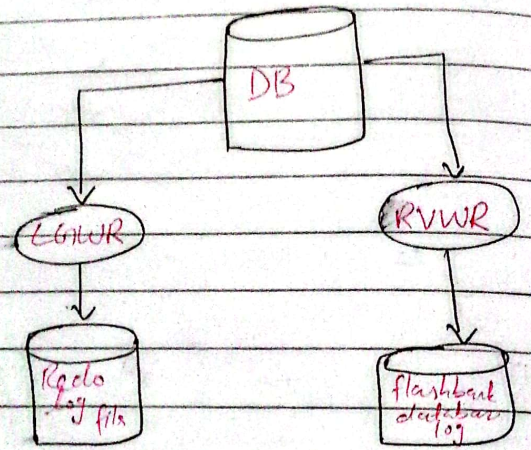
Flashback DB (introduced in 10.1)

Requires flash recovery area

⇒ Faster than traditional point in time recovery.

⇒ Flashback db is implemented using a new type of log file called the flashback database log.



maintains before image logs for block changes

Records are appended to flashback log.

uses RVWR background processes

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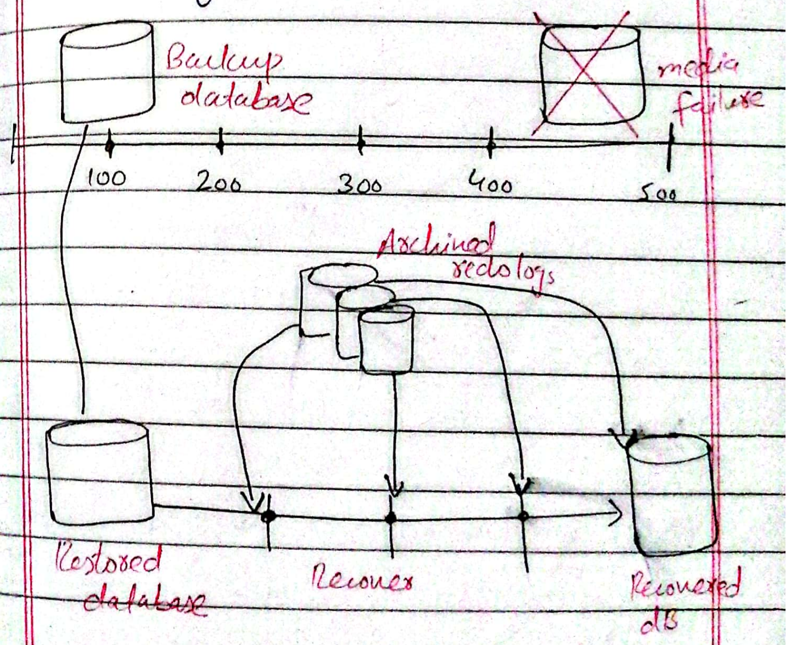
Difference between restoring and recovering?

Restoring:

⇒ Copying backup files from secondary storage (backup media) to disk.

Recovering:

⇒ Process of applying redo logs to the database to roll it forward.



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Q. How to size Shared pool?

⇒ Sizing Shared pool is critical task for optimizing performance.

⇒ Shared pool is part of SGA and proper sizing the shared pool helps in reducing parsing overhead, optimizing memory usage, and improving response times for queries.

⇒ Initial sizing of shared pool is based on system requirement.

- foo small system, size is 100m.
- foo large system, size is 200m.

⇒ Size of the shared pool can be set by using SHARED_POOL_SIZE parameter in the init.ora or spfile.

e.g

ALTER SYSTEM SET

SHARED_POOL_SIZE = 200M

⇒ V$SHARED_POOL_ADVICE provide advice on shared pool size based on current workload.

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⇒ Signs of improper shared-pool size:

- High parse time
- Library cache misses
- ORA - 04031 errors

⇒ Use automatic shared memory management (ASMM) for automatically managed the shared pool size.

⇒ Use automatic database diagnostic monitor (ADDM) and AWR reports used for automatic tunning.

Q. How to size the buffer cache?

⇒ The size of the database buffer cache is controlled by the INIT.ORA parameter DB_BLOCK_BUFFERS, which specifies the number of db blocks that will be contained in the buffer cache.

⇒ Since this expressed as database blocks, the size of the database buffer cache

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is:

= DB_BLOCK_BUFFERS * DB_BLOCK_SIZE

⇒ Buffer cache is part of SGA and it stores copies of data blocks that are read from datafiles.

⇒ Proper sizing ensures that frequently accessed data blocks remain in memory, reducing the need for disk I/O and improving overall database performance

⇒ Performane of buffer cache is monitored by calulating hit ratio

⇒ Use $VDB_CACHE_ADVICE to get insights from buffer cache dynamically.

⇒ Based on performance adjusted using:

DB_CACHE_SIZE